Xujiang Tang

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Visiting Student, Johns Hopkins University
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RESEARCH INTERESTS

Statistical learning theory, out-of-distribution generalization, learnability, optimization, stochastic processes, dynamical systems, operator-theoretic methods, causal inference, quantitative finance, and mathematical foundations of machine learning.

EDUCATION

Yangtze University College of Arts and Sciences, Jingzhou, China
B.Sc. in Mathematics and Applied Mathematics, expected June 2027
- Selected coursework: mathematical analysis, linear algebra, probability and statistics, differential equations, computational modeling, optimization theory, numerical computation, data structures, Python, R, and MATLAB.
- Selected marks: Optimization Theory 97/100; Numerical Computation Methods 96/100.

RESEARCH EXPERIENCE

Visiting Student, Johns Hopkins University, 2025-present
- Research on out-of-distribution generalization and learning theory for deep autoregressive models.

Theoretical Machine Learning and Optimization, 2024-present
- Work on learnability, generalization guarantees, and optimization-theoretic aspects of learning algorithms.

Koopman Operator Methods and Dynamical Systems, 2024-present
- Research on operator-theoretic approaches to long-sequence modeling and chaotic dynamics.

Rough Volatility and Quantitative Finance, 2023-present
- Research on stochastic-process models, MCMC-based simulation, and robust financial decision problems.

Spatial Transcriptomics and Biomedical Data Science, 2024-2025
- Work on single-cell and spatial data analysis from geometric and statistical perspectives.

PUBLICATIONS

Published and accepted papers
- Q. Li and X. Tang. HKGEduRec: A Knowledge Graph-Enhanced Dynamic Hybrid Framework for Educational Recommendation with Cold-Start Mitigation. IEEE ICMEIM 2025. EI-indexed.
- X. Tang and Q. Li. Logical Gene Encoding: A Bio-Inspired Approach for Energy-Efficient Automated Reasoning. IEIT 2025. EI-indexed. DOI: 10.2991/978-94-6463-803-5_81.
- Q. Li and X. Tang. Robust Optimal Reinsurance and Investment with Inflation Risk: A Game-Theoretic Approach and Explicit Solutions. AIMS Mathematics. Accepted, 2025.

Manuscripts under review
- X. Tang and C. Fan. Out-of-Distribution Generalization Error Bounds for Successful Prediction of Deep Autoregressive Algorithms. Submitted to ICML 2026.
- X. Tang. Quartic Difficulty: Assessing the Learnability of Unsupervised Learning Algorithms. Submitted to COLT.

HONORS AND ACTIVITIES

- Kaggle Silver Medal, Jigsaw Toxic Comment Classification Challenge.
- Regional Gold Medal, WorldQuant International Quant Championship.
- Third Prize, UCAS Graduate AI Forum.
- Multiple national second prizes in mathematical modeling competitions.
- Reviewer, ICLR 2026 Workshop.

PROFESSIONAL EXPERIENCE

Peking University - Open-Source LLM Group, Jun.-Aug. 2024
- Contributed to LLM deployment and retrieval-augmented generation systems.

Deloitte, Jul.-Sep. 2024
- Professional experience in consulting workflows, teamwork, and communication.

Guolian Securities, Jul.-Sep. 2024
- Applied statistical methods to securities research and quantitative workflows.

SKILLS

- Mathematics: optimization, numerical analysis, stochastic processes, differential equations, causal inference, statistical modeling.
- Programming and tools: Python, PyTorch, NumPy, Pandas, MATLAB, R, C++, LaTeX, Git/GitHub, Linux.
- Languages: Mandarin Chinese, English.